

Biopolymer enhanced vegetation establishment and their coupled effects on soil hydro-mechanical properties

Jiayu Gu^a, Junjun Ni^{a,*}, Yanbo Chen^b, Yufeng Gao^c

^a School of Transportation, Southeast University, Nanjing 211189, China

^b Department of Civil Engineering, Zhejiang University, Hangzhou 310058, China

^c Key Laboratory of Ministry of Education for Geomechanics and Embankment Engineering, Hohai University, Nanjing 210098, China

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ABSTRACT

Biopolymer amendments can modify soil structure and water dynamics, thereby influencing plant establishment and soil functional performance. This study examined the combined effects of three biopolymers (xanthan gum, guar gum, and chitosan) and three grasses—bermudagrass (*Cynodon dactylon*), tall fescue (*Festuca arundinacea*), and ryegrass (*Lolium perenne*)—on vegetation establishment and soil hydro-mechanical behavior. Cultivation tests showed that tall fescue exhibited the greatest adaptability among tested species, while xanthan had the least inhibitory effect on early growth, increasing final germination rate and shoot height by 10.0% and 10.3%, respectively, relative to unamended soil. Biopolymer addition markedly enhanced soil water retention, raising saturated water content from 25.7% to 30.8%, while xanthan-tall fescue (X-T) treatment further increased it to 32.2%. At a matric suction of 100 kPa, the saturation degree of X-T reached 74.1%, more than three times that of unamended soil, indicating a substantially improved moisture environment for plant development. Xanthan gum greatly reduced saturated hydraulic conductivity (from 7.4×10^{-7} m/s to 3.4×10^{-9} m/s), whereas X-T treatment maintained a relatively high conductivity (1.4×10^{-8} m/s) due to root-induced preferential flow pathways. These biopolymer- and root-driven changes in soil structure and water transport translated into higher shear resistance under both saturated and unsaturated conditions, with the strongest improvements observed at elevated matric suction. The biopolymer bonding regulates pore-scale water retention while roots modify flow pathways and contribute mechanical anchorage. This coupling between biopolymers and vegetation provides a promising strategy for improving soil functional quality and plant establishment on disturbed or engineered slopes.

1. Introduction

Rapid global economic development has accelerated the construction of transportation, water conservancy, and energy infrastructure, leading to extensive earthworks, mountain excavation, and slope regrading (Cao et al., 2025; Peng et al., 2019). These activities have substantially increased the area of engineered slopes and exposed bare ground. At the same time, intensified mineral resource exploitation, particularly in open-pit mines and quarries, has generated large areas of artificially disturbed slopes through surface stripping and soil removal (Rupke et al., 2007). Such bare slopes lack protective surface cover and are therefore highly susceptible to weathering, runoff-induced surface degradation, and shallow structural instability. The resulting soil degradation commonly manifests as substrate stripping, rill and gully formation, accelerated soil loss, and progressive deterioration of soil

structure and ecosystem functions. These processes not only undermine slope durability but also threaten long-term environmental sustainability and landscape stability.

Conventional hard-engineering approaches to slope protection, such as shotcrete, grouted rubble, and cement-based coverings, primarily rely on forming a rigid, low-permeability surface layer to resist erosion and shallow failures (Luo et al., 2018). Although these measures can provide rapid physical protection, they are associated with high construction costs, substantial embodied carbon emissions, and uncertain long-term performance (Müller et al., 2014; National Academies Of Sciences, 2012). More critically from a soil and ecological perspective, the low porosity and permeability of cementitious materials severely restrict the exchange of water and air within the soil profile, inhibiting root development, microbial activity, and natural soil formation processes (Li et al., 2019). Consequently, such treatments often eliminate pre-existing

* Corresponding author.

E-mail address: nijunjun@seu.edu.cn (J. Ni).

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ecological functions and conflict with the principles of sustainable land restoration.

Vegetation-based slope protection (bioengineering) has therefore gained increasing attention as a more sustainable alternative. Plant canopies intercept rainfall and reduce the kinetic energy of surface runoff, while root systems modify soil structure, enhance aggregate stability, and provide mechanical reinforcement (Gonzalez-Ollauri and Mickovski, 2017; Ng et al., 2020). Herbaceous roots mainly stabilize the shallow soil layer and improve near-surface stability, whereas deeper woody roots contribute to subsurface anchorage and improved slope integrity (Hao et al., 2023; Liu et al., 2019, 2025). Through these combined hydraulic and mechanical effects, vegetation can regulate soil moisture dynamics, maintain structural integrity, and enhance overall soil resilience. Despite these advantages, the practical implementation of vegetation-based slope protection on highly disturbed or barren slopes remains challenging. Such substrates typically contain limited organic matter, poor nutrient availability, and weak soil structure, resulting in low water-holding capacity and rapid surface runoff. These conditions hinder seed germination, reduce seedling survival, and slow vegetation establishment (Latocha et al., 2016; Wei et al., 2021). Moreover, during the early stages of plant growth, when a dense root network and sufficient canopy cover are developed, the soil surface remains highly vulnerable to disintegration (Wang et al., 2022). This temporal gap between planting and effective protection represents a major limitation of current bioengineering practices and highlights the need for complementary soil amendments that can rapidly improve site conditions and facilitate vegetation establishment.

Biopolymers are naturally derived, biodegradable polymers produced by plants, animals, or microorganisms, and have recently emerged as promising soil amendments in geotechnical and environmental engineering (Gu et al., 2025b). Owing to their adhesive and gel-forming properties, biopolymers can bind soil particles, enhance aggregate stability, and modify pore structure, thereby improving both mechanical strength and hydraulic behavior (Guo et al., 2025). Common examples such as xanthan gum, chitosan and β -glucan have demonstrated strong capabilities in promoting soil aggregation, increasing shear strength, and improving water retention (Gu et al., 2025b; Wang et al., 2023). These improvements are beneficial for maintaining near-surface soil integrity and hydro-mechanical stability, which are important for ecological slope restoration. The unconfined compressive strength (UCS) of sand amended with agar and starch can reach 487 kPa (Khatami and O Kelly, 2013), while clay amended with 0.5% glucan exhibits a UCS substantially higher than that of clay stabilized with 10% cement (Chang and Cho, 2014). In addition, biopolymer amendments can alter clay fabric and promote aggregation, which may influence hydraulic conductivity and moisture distribution in mixed soils (Biju and Arnepalli, 2020; Chang et al., 2015). These attributes make biopolymers attractive candidates for supporting vegetation-based slope protection systems. However, existing studies have largely focused on single biopolymer-single plant combinations, limiting our understanding of how different biopolymers interact with diverse plant species (Ni et al., 2023; Wan et al., 2024). Systematic comparisons across multiple biopolymers and vegetation types remain scarce. Furthermore, both biopolymers and plant roots strongly influence soil hydraulic properties, yet their combined effects on evaporation dynamics, water retention, and matric suction are not fully understood.

To address these gaps, this study investigates how different biopolymer amendments affect slope vegetation establishment and how biopolymer-plant interactions modify soil hydraulic and mechanical properties under varying matric suction conditions. A series of cultivation experiments was conducted to identify optimal biopolymer-vegetation combinations, followed by systematic evaluation of evaporation behavior, water retention, saturated hydraulic conductivity, disintegration resistance, and shear strength. The objective is to elucidate the suction-dependent, multi-scale reinforcement mechanisms of biopolymer-vegetation composite soils and to provide a scientific basis

for improving near-surface soil stability and hydro-mechanical performance in ecological slope restoration.

2. Materials and methods

2.1. Materials

2.1.1. Soil

The soil used in this study was collected from a natural slope in Jiangning District, Nanjing, China. To ensure consistency in the test material, the sampled soil was oven-dried at 40°C, gently crushed, and sieved through a 2 mm standard mesh prior to testing. Particle-size distribution was determined following the guidelines of ASTM D7928–21 (2021a), and the resulting grain-size curve is presented in Fig. 1. The proportions of sand, silt, and clay were determined to be 16.7%, 68.1%, and 15.2%, respectively. The Atterberg limits and standard compaction characteristics were obtained in accordance with ASTM D4318–17 (2018) and ASTM D698–12 (2021b). Based on the Standard Practice for Classification of Soils for Engineering Purposes as specified in ASTM D2487–17 (2020), the soil is classified as silty clay (CL). The other detailed physical and index properties of the tested soil are summarized in Table 1. The soil contains a high proportion of silt and a relatively low clay fraction, which limits aggregate stability and reduces water retention. Consequently, it is highly susceptible to structural breakdown and erosion under rainfall or surface runoff (Martínez-Murillo et al., 2013).

2.1.2. Biopolymer

Three types of biopolymers—chitosan, guar gum, and xanthan gum—were selected in this study due to their wide applicability in soil improvement, slope stabilization, and environmental remediation. These biopolymers are natural, biodegradable, and environmentally benign materials that can enhance soil structure, water retention, and mechanical strength through physicochemical interactions with soil particles.

Chitosan is a cationic polysaccharide derived from the deacetylation of chitin, which is abundant in the exoskeletons of crustaceans. It features high solubility, strong adsorption capacity, and excellent biodegradability. The positively charged amino groups along its molecular chains enable electrostatic interactions with negatively charged soil particles, thereby promoting the formation of water-stable aggregates and enhancing soil structure (Hataf et al., 2018). Guar gum, a nonionic or weakly anionic polysaccharide extracted from legume seeds, exhibits high viscosity and strong water retention ability (Mudgil et al., 2014).

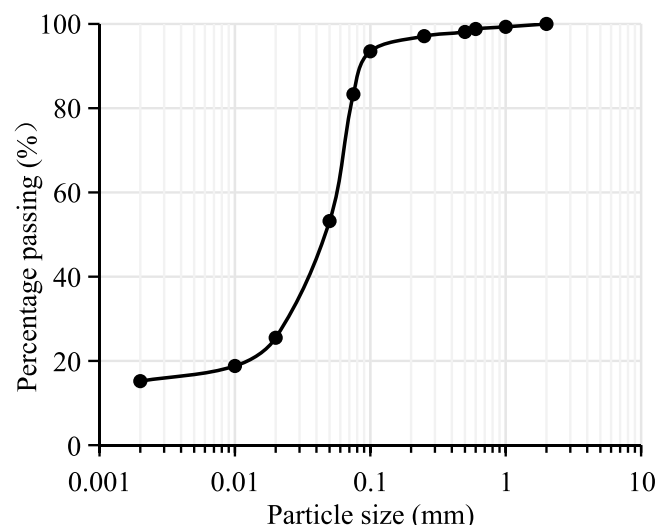


Fig. 1. Particle size distribution curve of soil.

Table 1
Physical properties of soil.

Specific gravity	Plastic limit (%)	Liquid limit (%)	Optimum water content (%)	Maximum dry density (kg/m ³)
2.69	16.5	31.7	19.8	1770

The abundant hydroxyl groups along its molecular chains promote the formation of a dense hydrogen-bonding network within the soil matrix. This network enhances the water retention capacity of the soil and reduces evaporation losses. Meanwhile, it enhances the contact between soil particles and seeds, facilitating seed anchorage within the substrate and improving resistance to detachment under rainfall-induced runoff. Xanthan gum is an anionic polysaccharide synthesized via microbial fermentation, most commonly by *Xanthomonas campestris* (Gu et al., 2025b). It is characterized by pseudoplastic (shear-thinning) behavior and excellent thickening and stabilizing properties. Owing to its high molecular weight and viscosity, xanthan gum can markedly increase the viscosity of topsoil slurry even at low concentrations. This property enhances the uniformity and adhesion of xanthan gel when it is applied via hydraulic spraying. Furthermore, the stable three-dimensional network formed by xanthan contributes to prolonged water retention and improved soil cohesion.

Chitosan, xanthan gum, and guar gum enhance soil structure and water retention through electrostatic interactions, hydrogen-bonding networks, and three-dimensional gel formation. These effects improve soil cohesion, seed-soil contact, and resistance to erosion, making biopolymers suitable for slope stabilization and vegetation establishment. The molecular structures of chitosan, xanthan gum and guar gum are shown in Fig. 2.

2.1.3. Vegetation

On bare slopes, where soil conditions are poor and growth substrate is limited, the selection of suitable vegetation is critical for ecological restoration and slope stabilization. Fast-growing species with high

survival rates were prioritized to achieve effective ground cover and erosion control. Three herbaceous species commonly used for slope protection in southern China were selected: bermudagrass (*Cynodon dactylon*), tall fescue (*Festuca arundinacea*), and ryegrass (*Lolium perenne*).

Bermudagrass is a warm-season turfgrass with a well-developed and extensive root system. It exhibits strong drought tolerance, adaptability to nutrient-poor soils, and high mechanical resistance, allowing it to form a dense turf suitable for rapid slope coverage. Tall fescue is a cool-season turfgrass with a deep and compact fibrous root system. It shows good tolerance to shade, high soil moisture, and a wide range of environmental conditions, and is commonly used as a perennial foundational species in southern regions to provide sustained soil stabilization. Ryegrass is a cool-season herbaceous grass known for rapid germination and fast growth. Its root system is relatively shallow, making it suitable as a temporary or winter cover to quickly establish ground cover while other perennial species mature.

2.2. Sample preparation

In this study, the biopolymer dosage was fixed at 0.5% of dry soil mass. This dosage was selected as a low-to-moderate and practically relevant level. Previous studies have shown that biopolymer contents within this range can already improve soil water retention and mechanical strength. At the same time, this dosage helps avoid excessive slurry viscosity, poor mixing uniformity, and severe inhibition of seed germination (Badakhshan et al., 2023; Chang et al., 2020). Chitosan, xanthan gum, and guar gum differ in molecular structure and ionic characteristics. However, a unified dosage was adopted to provide a consistent basis for comparing their effects on vegetation establishment and subsequent hydro-mechanical responses. The aim at this stage was comparative screening under the same amendment level, rather than dosage optimization for each individual biopolymer. The oven-dried soil was first blended with biopolymer powder at a mass ratio of 100:0.5 and thoroughly mixed to ensure uniform dispersion of the biopolymer

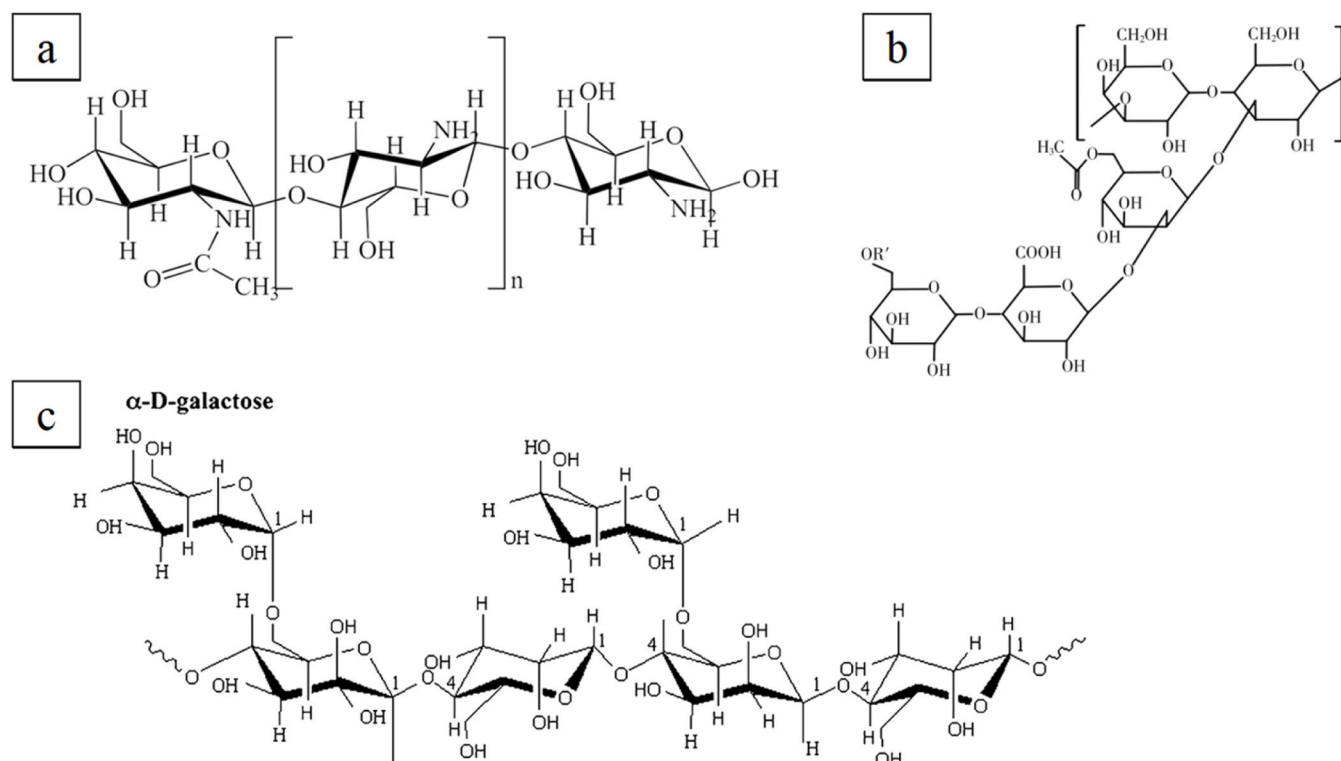


Fig. 2. Structure formulas of (a) chitosan (Islam et al., 2017), (b) xanthan gum (Gu et al., 2025b) and (c) guar gum (Mudgil et al., 2014).

throughout the soil matrix. Distilled water was then gradually added at 19.8% of the dry soil mass, corresponding to the optimum water content. Mixing was performed until the material showed a consistent color, a uniform distribution of moisture, and no discernible aggregates, which was taken as evidence of a homogeneous soil–water–biopolymer mixture.

Immediately thereafter, the prepared mixture was sealed in airtight polyethylene bags and stored in a controlled-temperature environment for 12 h to allow moisture redistribution and initial biopolymer hydration before compaction. This step was regarded as part of the specimen preparation procedure rather than a full curing stage. Following equilibration, the soil was compacted into cylindrical acrylic molds with an internal diameter of 70 mm and a height of 50 mm using a three-layer static compaction procedure. All specimens were prepared at a target dry density of 1.5 g/cm³. This value was selected to represent the relatively loose surface layer commonly used in ecological slope restoration. The same density-controlled preparation method was applied to all treatments. This ensured comparable initial compactness. The use of a unified dry density also minimized density variation among treatments. Otherwise, it would be difficult to distinguish whether the measured changes in hydraulic and mechanical behavior were caused primarily by biopolymer amendment or by differences in specimen density.

Prior to seeding, bermudagrass, tall fescue, and ryegrass seeds were screened using the water-floatation method to remove lightweight, damaged, or underdeveloped seeds, thereby ensuring high germination potential. A total of 80 selected seeds were uniformly distributed over the compacted soil surface (corresponding to a seeding density of about 20,000 seeds/m²). This relatively high seeding density was adopted to promote rapid establishment on compacted soil (De Baets et al., 2007). A thin layer of loose soil (approximately 2 mm) was then applied to cover the seeds to maintain moisture and reduce seed relocation during early watering. After seeding, all specimens were transferred to an artificial climate chamber for controlled cultivation. The environmental conditions were controlled at a temperature of 20°C, relative humidity of 80%, and a photoperiod of 16 h/day. Water was supplied periodically using a calibrated fine-spray system to maintain soil moisture near the optimum or the predetermined cultivation moisture level. Under these controlled conditions, additional hydration of the biopolymer may have continued during the cultivation stage. Vegetation characteristics, including germination rate and shoot height, were monitored throughout the cultivation period. These two indicators were used to evaluate vegetation establishment and early growth efficiency because they directly reflect seed emergence, seedling survival, and above-ground development in the tested soil matrix. At the plant-screening stage, the primary objective was to identify the grass species most suitable for establishment in biopolymer-amended soil. Therefore, germination rate and shoot height were considered sufficient for evaluating plant establishment efficiency, and additional root morphological measurements were not required for this screening purpose. After the hydraulic and mechanical tests, the roots were carefully separated from the soil by washing. The recovered roots were oven-dried to constant mass and weighed.

2.3. Laboratory tests

To investigate the effects of biopolymers on plant establishment and their coupled influence on soil hydro-mechanical behavior, a series of laboratory tests were conducted. The overall test plan is summarized in Table 2. Three categories of tests were performed to evaluate soil evaporation, water retention, hydraulic conductivity, disintegration resistance, and shear strength under the combined action of vegetation and biopolymers. The experimental apparatus for each test is shown in Fig. 3, and the detailed procedures are described in the following sections.

Table 2
Detailed test plan of this study.

Test series	Biopolymer type	Vegetation species	Test Parameters
Vegetation growth tests	Xanthan gum Guar gum Chitosan Controlled	Bermudagrass Tall fescue Ryegrass	Vegetation germination rate and shoot height
Hydraulic property tests	Xanthan	Tall fescue	Evaporation Water retention Saturated hydraulic conductivity Disintegration resistance
Mechanical property tests	Xanthan	Tall fescue	Shear strength under different matric suction

2.3.1. Evaporation characteristics test

Prior to the evaporation test, the vegetated soil specimens were saturated via a capillary rise method (or bottom-up wetting approach) to minimize air entrapment while preserving soil structure (Klute, 1986). To prevent root hypoxia (suffocation) caused by full inundation, a controlled bottom-watering protocol was adopted (Leung et al., 2015). The soil containers were placed in shallow trays containing a small quantity of water, allowing moisture to permeate the soil matrix and root network through capillary action. The water level in the tray was incrementally raised, and the process was terminated once the soil surface appeared visibly moist. This ensured that the specimens reached a state of capillary saturation while the uppermost soil layer remained sufficiently aerated for root respiration.

After saturation, excess water in the tray was removed, and the initial saturated mass of each specimen was recorded. The specimens were then transferred to a climate chamber (Fig. 3(a)) at 25°C and 40% relative humidity. These controlled conditions imposed a constant atmospheric demand, enabling accurate determination of evaporation rates. Mass of specimens was recorded at regular intervals using a high-precision electronic balance. The measured mass loss over time was used to calculate the evaporation rate for each treatment.

2.3.2. Water retention test

To quantify the effects of biopolymer amendment and vegetation on soil water retention, a series of tests were conducted following the standard methods for determining the soil water characteristic curve (ASTM, 2016a). To eliminate the direct interference of plant transpiration, the above-ground biomass was entirely removed prior to testing. Undisturbed soil–vegetation samples (61.8 mm in diameter and 40 mm in height) were collected using a standard cutting ring. The samples were fully saturated under vacuum for 24 h (Fig. 3(d)). After saturation, the specimens were weighed and placed in the pressure membrane apparatus (Fig. 3(b)). Pore water was extracted by applying incremental air pressures and allowing it to drain through the porous ceramic plate. The expelled water was collected in a burette until no further drainage occurred, indicating that equilibrium had been reached. Soil water content at each matric suction was calculated from the mass change. The apparatus features low- and high-pressure controllers. Suctions below 200 kPa were applied using the low-pressure controller, whereas higher suction values were applied after switching to the high-pressure controller. This dual-range operation ensures accurate determination of the water retention curve across the full suction spectrum.

2.3.3. Saturated hydraulic conductivity test

The saturated hydraulic conductivity (k_s) was evaluated using the falling-head permeability test according to ASTM D5856 (ASTM, 2024). For vegetated specimens, the above-ground parts were removed before testing, whereas the root systems were retained intact within the soil matrix. This treatment was adopted because the objective was to characterize the saturated hydraulic conductivity of the

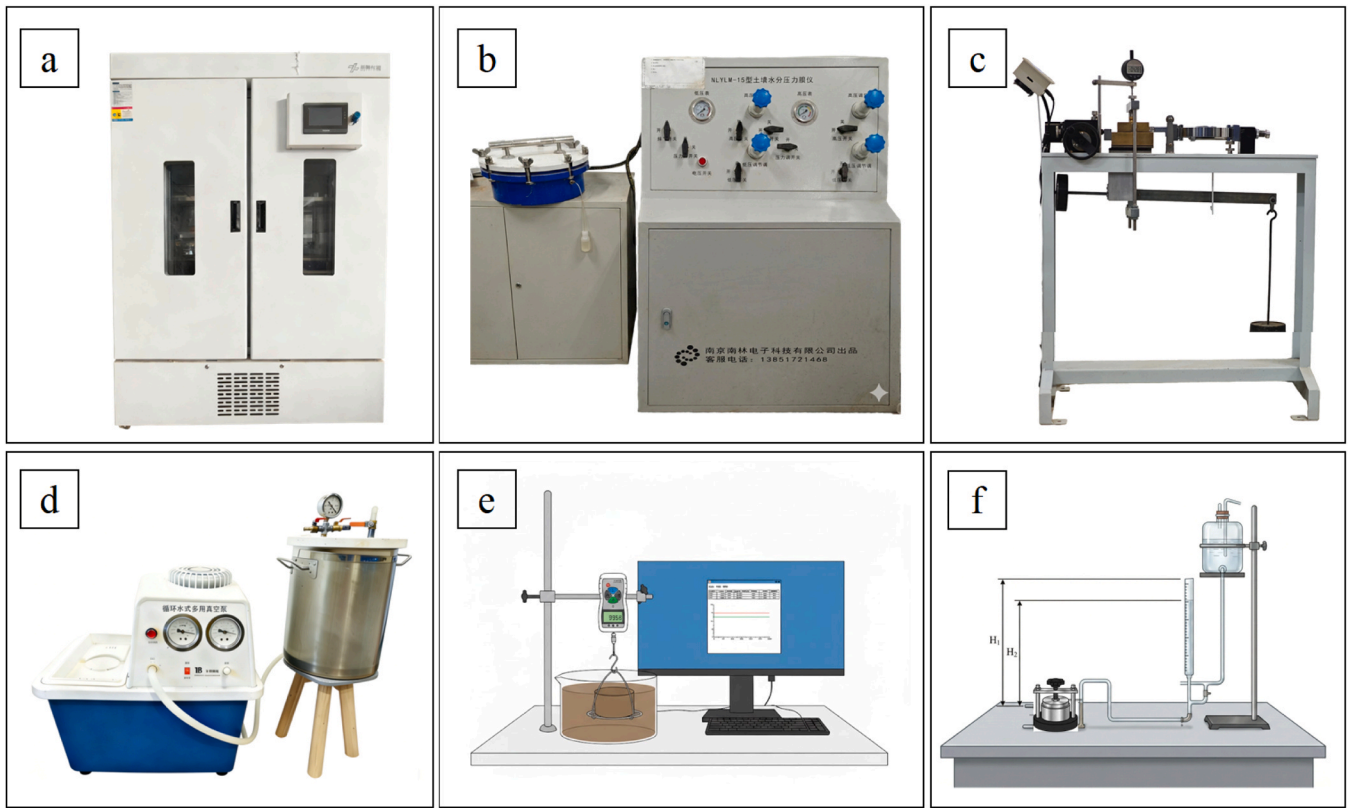


Fig. 3. The devices used in this study (a) Artificial climate growth chamber; (b) Pressure membrane apparatus; (c) Direct shear apparatus; (d) Vacuum pump; (e) Slake durability apparatus; (f) Falling-head permeability apparatus.

biopolymer-vegetation composite soil as a whole. It should be noted that intact roots may influence k_s in two opposite ways: root occupation and rhizosphere products can reduce pore connectivity and increase flow tortuosity, while root channels may provide localized preferential flow pathways (Cui et al., 2025). The measured k_s therefore represents the net hydraulic response of the vegetated soil system. A schematic of the experimental setup is shown in Fig. 3(f). Vacuum-saturated specimens, prepared as described in Section 2.3.2, were carefully placed into the permeameter cell. To prevent preferential flow along the soil-wall interface, a thin layer of petroleum jelly was applied to the inner wall of the mold prior to installation. De-aired water was introduced to the system to flush the tubing and ensure the removal of any residual entrapped air. Once a continuous and steady-state outflow was established, the standpipe was filled to a specific initial hydraulic head (h_1). The test commenced by opening the valve and recording the time interval (t) required for the water level to drop to a designated final head (h_2). The k_s value was subsequently calculated based on Darcy's law using Eq. (1).

$$k_s = \frac{aL}{At} \ln\left(\frac{h_1}{h_2}\right) \quad (1)$$

where a and A represent the cross-sectional areas of the standpipe and the soil specimen, respectively, and L denotes the specimen length. This testing protocol provided quantitative insights into the modification of soil water flow characteristics induced by the biopolymer treatment and vegetation root network.

2.3.4. Disintegration resistance test in still water

The disintegration resistance of the biopolymer-vegetation soil was assessed using a static disintegration test modified from ASTM D4644 (ASTM, 2016b). The test setup consisted of a wire-mesh basket (aperture: 2 mm) suspended from a high-precision electronic balance above a

water tank filled with distilled water, as shown in Fig. 3(e). Dried cylindrical specimens were placed at the center of the mesh basket and gently immersed in static water. Mass loss resulting from slaking and particle detachment was recorded continuously. The residual mass was measured at designated time intervals (e.g., 1, 5, 10, and 20 min) until disintegration stabilized or the specimen completely failed. The disintegration ratio (D_r) was then computed using Eq. (2) to quantify the water stability of the soil aggregates.

$$D_r = \frac{M_{\text{initial}} - M_{\text{residual}}}{M_{\text{initial}}} \times 100\% \quad (2)$$

where M_{initial} is the initial dry mass of the specimen, and M_{residual} is the retained mass in the basket at a given time.

2.3.5. Suction-controlled direct shear test

Shear strength is a key parameter for evaluating slope stability, but it is highly sensitive to soil moisture conditions. To represent two contrasting hydrological states relevant to near-surface slope soils, direct shear tests were conducted at matric suctions of 0 and 200 kPa. The suction of 0 kPa represents the saturated condition, whereas 200 kPa represents a relatively dry unsaturated state close to the residual water content range indicated by the SWRC results. Saturated specimens were prepared by vacuum saturation (Fig. 3d), and unsaturated specimens were conditioned to 200 kPa using a pressure plate apparatus (Fig. 3b). The specimens were first equilibrated to the target suction and then sheared under constant water content conditions. Suction was not actively monitored during shearing, but was assumed to remain approximately constant because of the short test duration and sealed testing condition. Maintaining constant suction is essential for capturing the shear strength associated with the prepared hydrological states, which is particularly relevant for assessing short-term slope stability under rapid wetting or loading. The tests were carried out using a direct

shear apparatus (Fig. 3(c)). Following geotechnical testing guidelines (ASTM, 2017), a shear rate of 1.0 mm/min was adopted, which falls within the recommended range for quick shear tests (0.8–1.2 mm/min). Shear stress and horizontal displacement were recorded continuously, and testing was terminated when a peak followed by a stabilized shear stress was observed, or when 6 mm of shear displacement (approximately 10% shear strain) was reached.

3. Results and discussion

3.1. Seed germination and growth performance

Seed germination rate and shoot height were used as primary indicators of plant establishment and early growth under different soil amendment conditions. Germination reflects the suitability of the soil environment for seed activation and emergence, whereas shoot height provides an integrated measure of early biomass development and the potential for canopy formation that can influence surface hydrological processes such as rainfall interception and runoff attenuation. Figs. 4 and 5 present the temporal evolution of germination and shoot height for the three tested species, while representative photographs of plant development at different cultivation stages are shown in Fig. 6. Germination was monitored for the first 15 days, as stabilization occurred after approximately 10 days, whereas shoot height was recorded over the full 30-day cultivation period.

Fig. 4 shows that bermudagrass exhibited the slowest germination among the three species, with no visible emergence before Day 4 in unamended soil. In contrast, ryegrass and tall fescue germinated more rapidly, reaching 11.3% and 28.8% germination by Day 2, respectively. The incorporation of biopolymers generally delayed early germination across all species, although the magnitude of this effect varied with polymer type. Chitosan produced the most pronounced retardation, followed by guar gum, whereas xanthan gum exerted the weakest inhibitory influence. With continued cultivation, germination rates increased steadily in all treatments. Stabilization occurred around Day 8 in unamended soil, while biopolymer amended soils required approximately four additional days to reach a stable level, suggesting that soil amendments primarily influenced the early establishment phase rather than final germination capacity.

Shoot height development (Fig. 5) followed patterns broadly consistent with germination trends. During the early growth stage (e.g., Day 8), plants in unamended soil exhibited faster shoot elongation than those in amended soils. At this stage, shoot height in unamended soil exceeded that in chitosan-, xanthan-, and guar gum-amended soils by

20.6–31.5%, 0.8–7.1%, and 8.9–17.1%, respectively. Among the biopolymers, xanthan exerted the least constraint on early shoot elongation. After approximately 20 days, shoot growth progressively slowed and approached a plateau across all treatments, indicating that biopolymer effects were concentrated mainly during initial establishment rather than later vegetative development.

Fig. 7 summarizes final germination rates and ultimate shoot heights under different biopolymer treatments. Tall fescue consistently achieved the highest germination rate among the three species, whereas bermudagrass and ryegrass exhibited comparable values. Relative to bermudagrass, tall fescue showed higher germination by 6.8%, 16.7%, 14.6%, and 18.6% in unamended, chitosan-, xanthan-, and guar-gum-amended soils, respectively. In terms of shoot height, bermudagrass produced the lowest values across all treatments, while tall fescue displayed a clear growth advantage, achieving final heights 28.0–41.8% greater than bermudagrass and 9.1–22.7% greater than ryegrass. Considering both germination performance and biomass accumulation, tall fescue was identified as the most suitable species for vegetation establishment on amended slopes. For tall fescue, biopolymer effects varied distinctly by polymer type. Compared with unamended soil, final germination rate changed by −8.8%, +10.0%, and +4.2% following chitosan, xanthan gum, and guar gum addition, respectively. Correspondingly, final shoot height changed by −3.2%, +10.3%, and +3.8%. These results indicate that xanthan gum provided the most favorable soil environment for tall fescue establishment and early growth. Consequently, the subsequent investigation into the hydro-mechanical properties of biopolymer-vegetation amended soil will focus specifically on the xanthan gum and tall fescue. The total root biomass of the X-T treatment was 2.85 g dry mass per specimen, which was higher than that of U-T (2.34 g). This result is consistent with the higher germination rate and greater shoot height observed in the X-T treatment, indicating that xanthan gum amendment promoted not only above-ground growth but also overall root development of tall fescue. The higher total root biomass further suggests a stronger potential influence of the root system on soil structure, water transport pathways, and mechanical reinforcement.

The generally lower germination rate observed in biopolymer-amended soils relative to unamended soil can be attributed to two interacting mechanisms. First, unamended soil typically possesses higher porosity and better gas exchange, facilitating oxygen diffusion to germinating seeds and supporting radicle emergence. In contrast, biopolymer incorporation tends to reduce total porosity and partially obstruct pore connectivity, limiting soil aeration and seed respiration and thereby slowing germination (Ayeldeen et al., 2017). Second, certain biopolymers, particularly seminatural macromolecules such as

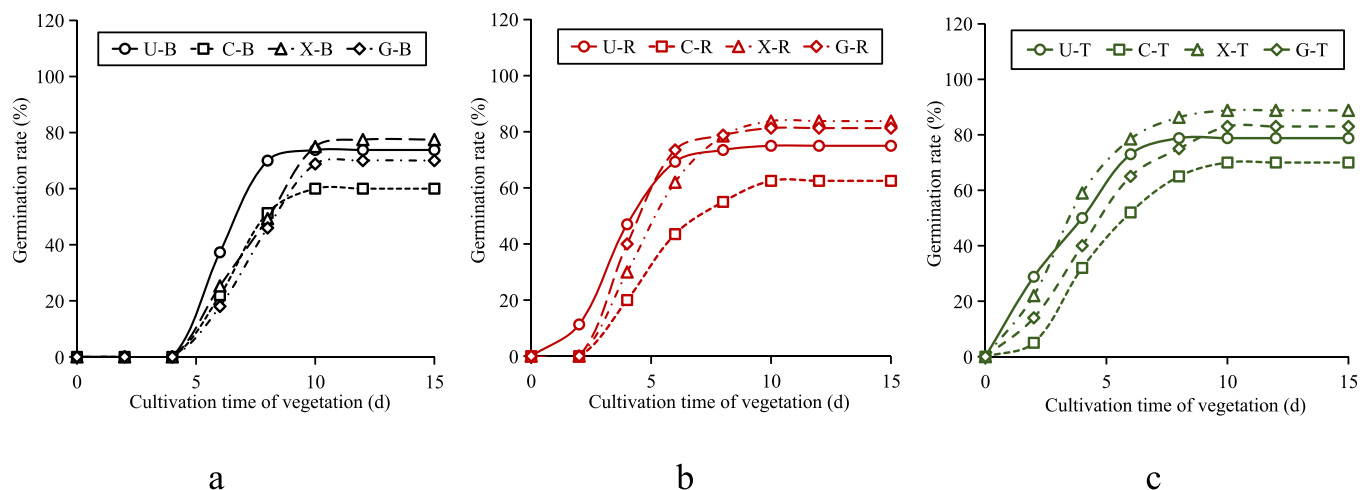


Fig. 4. Germination rate of (a) bermudagrass; (b) ryegrass; (c) tall fescue. **Note:** U = unamended soil; X = xanthan gum amended soil; G = guar gum amended soil; C = chitosan amended soil. B = bermudagrass; R = ryegrass; T = tall fescue. These abbreviations have been used in Figs. 5–7.

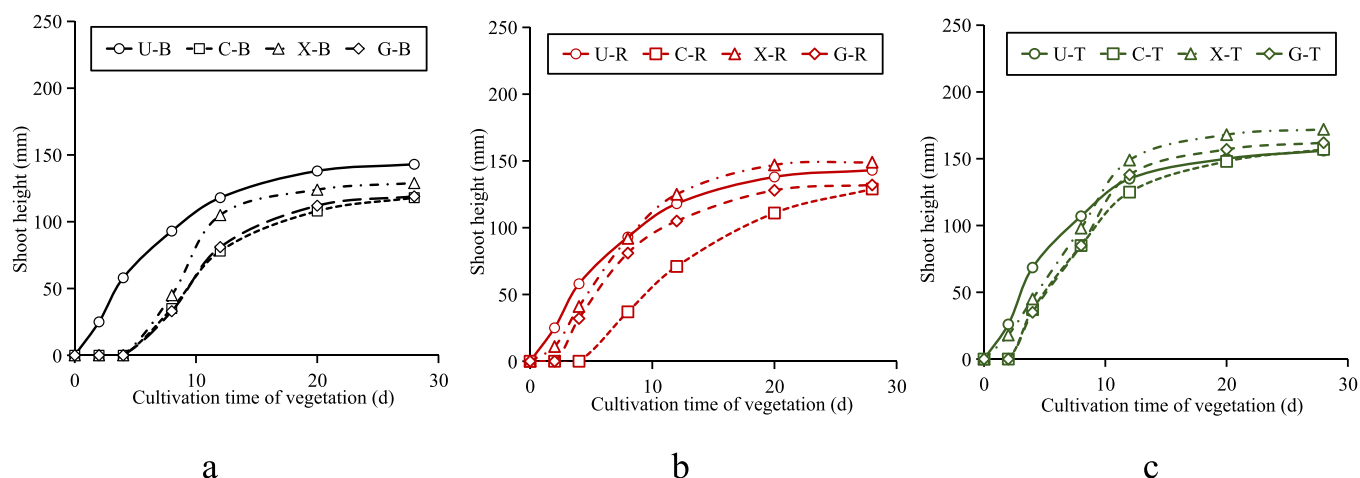


Fig. 5. Shoot height of (a) bermudagrass; (b) ryegrass; (c) tall fescue. Abbreviations are the same as those in Fig. 4.

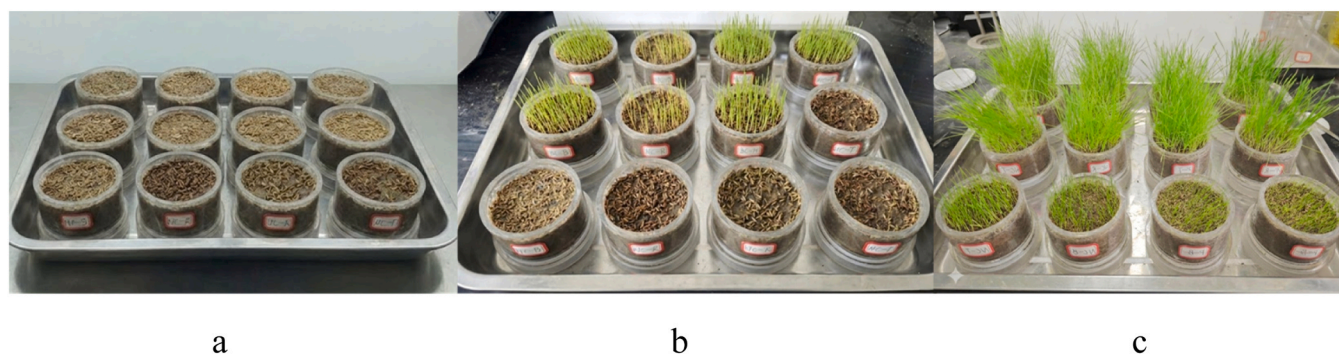


Fig. 6. Photographs of the plants at the cultivation time of (a) 0 day; (b) 4 days; (c) 7 days. **Note:** From left to right, the treatments correspond to U, X-T, G-T, and C-T, while from top to bottom, the species are T, R, and B. Abbreviations are the same as those in Fig. 4.

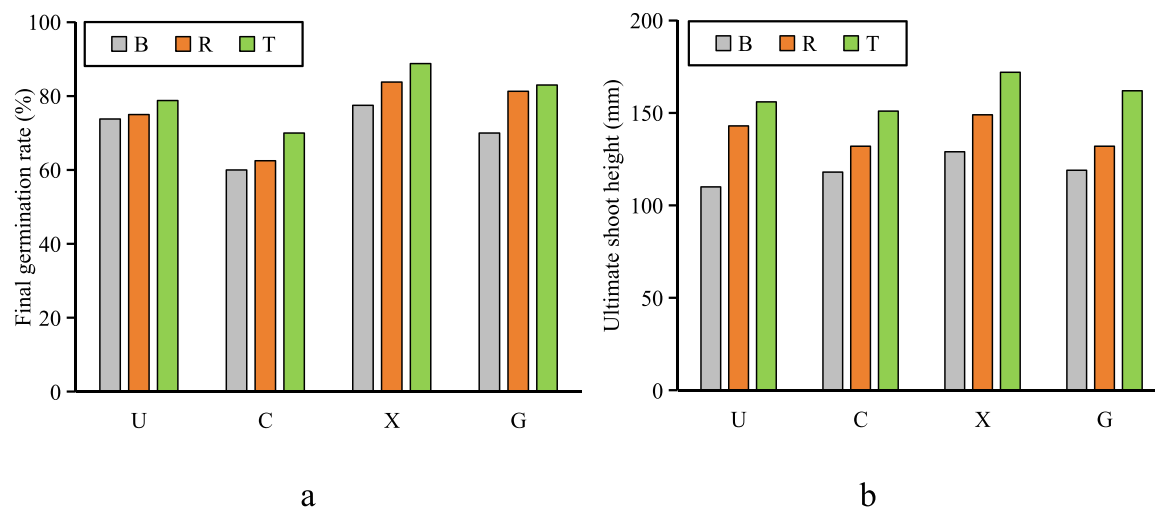


Fig. 7. (a) Final germination rates and (b) ultimate shoot height of the three species. Abbreviations are the same as those in Fig. 4.

chitosan, exhibit strong cohesive behavior that can promote the formation of a thin, consolidated surface crust (Orts et al., 2007). This mechanically resistant layer increases the force required for seedling emergence, potentially retarding shoot elongation and delaying the transition from germination to seedling establishment (Hillel, 2003).

3.2. Evaporation

As an open soil-water-air system, slope soil exhibits dynamic moisture behavior controlled by the interplay of rainfall, infiltration, and evaporation. Among these processes, evaporation dominates water loss and therefore serves as a critical indicator of soil hydraulic response. Figs. 8(a) and 8(b) present the temporal evolution of evaporation rates and cumulative water loss for the different treatments. As shown in

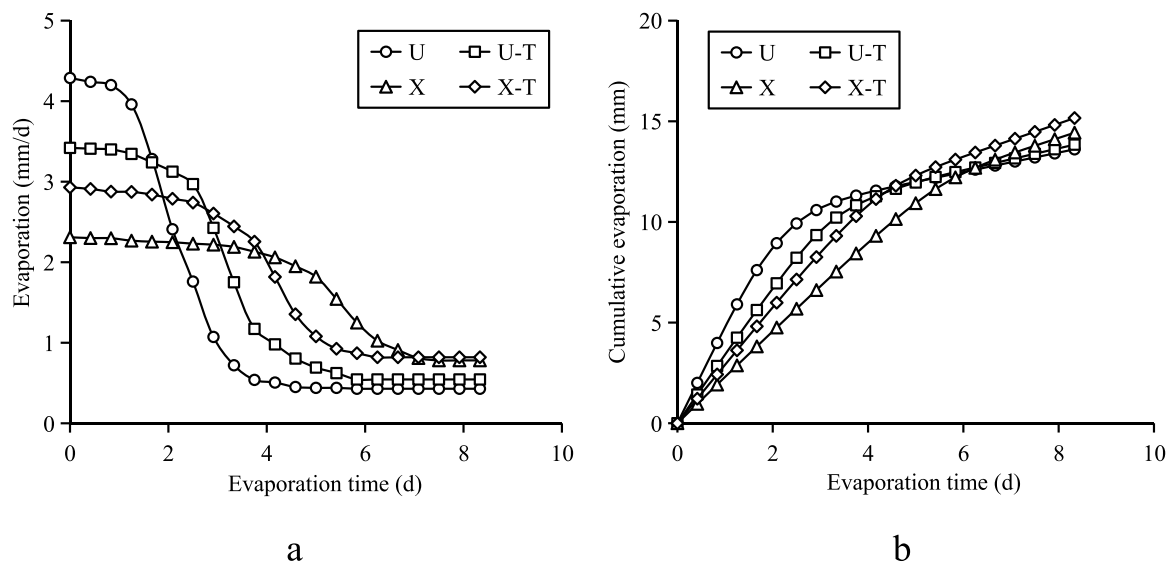


Fig. 8. (a) Evaporation rate and (b) cumulative water loss of untreated and xanthan-amended soils, with and without tall fescue vegetation. **Note:** U = unamended soil; X = xanthan amended soil; U-T = tall fescue only soil; X-T = xanthan-tall fescue composite soil. These abbreviations have been used in Figs. 9–15.

Fig. 8(a), soil evaporation followed the classical three-stage pattern: the constant rate stage (CRS), falling rate stage (FRS), and residual stage (RS). During the CRS, the unamended soil (U) exhibited the highest initial evaporation rate (4.29 mm/day), whereas the xanthan amended soil (X) showed the lowest value (2.31 mm/day). This substantial reduction indicates that xanthan gum effectively constrained capillary flow toward the surface and limited rapid liquid water loss in the early drying stage. The tall fescue only soil (U-T) and the xanthan-tall fescue composite soil (X-T) displayed intermediate initial rates, reflecting the competing effects of surface shading (which suppresses evaporation) and root water uptake (which enhances water depletion).

As evaporation progressed, the system transitioned into the FRS, where moisture transport was primarily controlled by soil hydraulic properties. The transition from CRS to FRS differed markedly among treatments. The U soil entered the FRS within approximately 2 days, whereas the X soil maintained the CRS for up to 5 days, demonstrating that biopolymer amendment significantly delayed the drying front migration and prolonged surface moisture availability. Compared with U, the U-T shifted more slowly into the FRS, confirming the protective role of vegetation cover in moderating surface drying. In contrast, the X-T entered the FRS earlier than X soil, suggesting that root water extraction partially counteracted the moisture-retaining effect of xanthan gum. Once in the FRS, evaporation rates declined progressively as liquid continuity in the pore network diminished. The fastest decline occurred in U (average reduction of 1.40 mm/day), followed by U-T (1.28 mm/day), X-T (0.79 mm/day), and X (0.54 mm/day). This sequence indicates that biopolymer treatment enhanced hydraulic continuity at low water contents, thereby sustaining moisture transport during drying. In the RS, evaporation stabilized at low values. The U exhibited the lowest residual rate (0.43 mm/day), reflecting limited water retention, whereas X and X-T maintained higher residual rates (0.78 and 0.82 mm/day, respectively), consistent with improved water storage due to polymer-induced microstructural modification.

Fig. 8(b) illustrates the cumulative evaporation behavior. During the early drying period, U showed the fastest water loss because of its high initial evaporation rate. After approximately 2 days, when U transitioned into the FRS, the increase in cumulative evaporation slowed markedly. In contrast, U-T and X-T continued to lose water at relatively high rates due to sustained transpiration-driven flux. By around Day 4, the cumulative evaporation of U-T and X-T exceeded that of U, while X remained lower due to its prolonged CRS. At the end of the experiment, X-T exhibited the highest cumulative evaporation, whereas U showed

the lowest. This pattern is primarily attributable to treatment-induced differences in saturated and residual water contents, as cumulative evaporation is governed by the difference between these two states. Both biopolymer and vegetation increased water storage capacity, with a more pronounced effect on saturated water content. Consequently, although biopolymer reduced short-term evaporation intensity, the combined biopolymer-vegetation system ultimately enabled greater total water release over the full drying cycle.

3.3. Water retention capacity

Water retention characteristics under varying matric suction are shown in Fig. 9. The soil water retention curves reveal distinct hydraulic responses among the four treatments, particularly in terms of saturated water content (w_{sat}), air-entry value (AEV) and residual water content. As illustrated in Fig. 9(a), both biopolymer amendment and vegetation establishment exerted a pronounced influence on w_{sat} and AEV. U exhibited the lowest w_{sat} of 25.7%, whereas xanthan amended soil (X) increased to 30.8% and tall fescue only soil (U-T) to 28.1%. This indicates that biopolymer addition was more effective than vegetation alone in enhancing soil water storage at saturation, likely due to polymer-induced pore modification and increased micro-scale water holding capacity. The combined biopolymer-vegetation treatment yielded the highest w_{sat} of 32.2%, demonstrating a cumulative effect of both components. However, this increase was less than the arithmetic sum of individual contributions, suggesting partial competition between biopolymer and root systems for pore space or heterogeneous distribution of water-retaining zones within the soil matrix. Consistent with w_{sat} trends, AEV increased from approximately 10 kPa in U soil to 30–50 kPa in amended soils, indicating greater resistance to desaturation and improved structural stability against air invasion. The higher AEV in X and X-T implies that biopolymer-induced pore constriction effectively delayed air penetration during drying.

With increasing matric suction, all soils exhibited progressive desaturation toward residual conditions, though the transition thresholds differed markedly. U reached residual water content at approximately 50 kPa, while U-T required around 100 kPa. In contrast, X and X-T did not approach residual conditions until suction exceeded 200 kPa, confirming that biopolymer treatment substantially extended the effective water retention range. Residual water content followed a pattern consistent with w_{sat} . U exhibited the lowest residual water content (2.7%), while vegetation treatment alone increased this value to 3.8%.

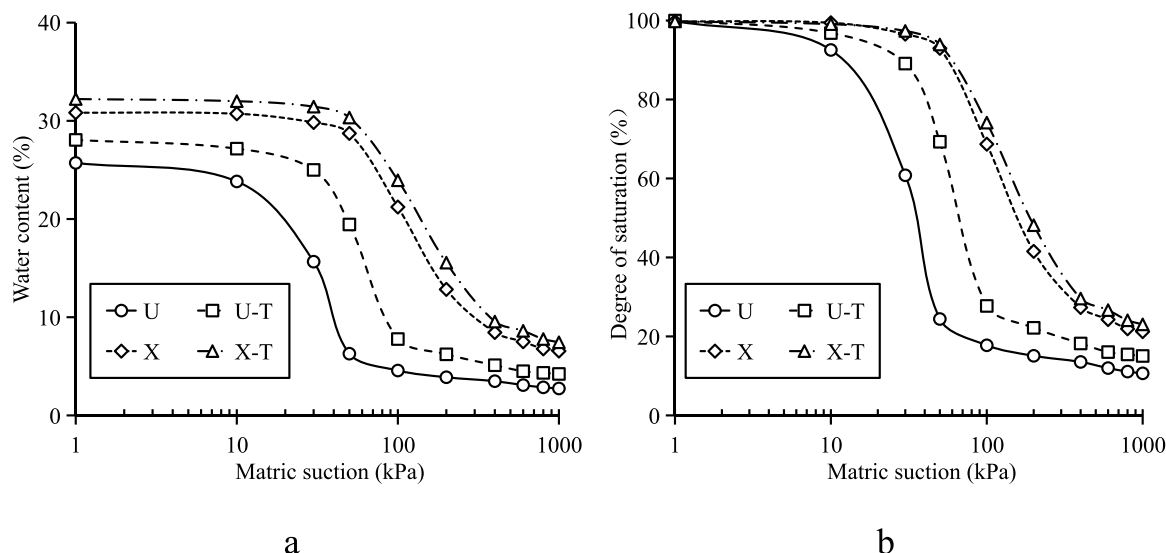


Fig. 9. (a) Water content and (b) degree of saturation as functions of matric suction for untreated and xanthan-amended soils, with and without tall fescue vegetation. Abbreviations are the same as those in Fig. 8.

Biopolymer amended soils (X and X-T) showed even higher residual water contents (6.6% and 7.4%, respectively). These results demonstrate that biopolymer amendment was the dominant factor controlling water retention at high suction levels. The increase in air-entry value indicates that air invasion into the pore network was delayed after biopolymer treatment. This can be attributed to biopolymer coatings and bridges that narrow effective pore throats and reduce the continuity of larger capillary pathways, so that a higher suction is required for air entry. By contrast, the increase in residual water content is more closely related to the hydrophilic nature of the polymer. The polymer chains provide additional adsorption sites and retain part of the pore water as weakly mobile or bound water, allowing water to persist even at relatively high suction (Gu et al., 2026). Therefore, the increase in air-entry value is mainly associated with pore narrowing, whereas the increase in residual water content is more closely related to adsorption and hydration along the polymer network. Similar interpretations have been reported for biopolymer-treated soils and in generalized soil-water retention theory distinguishing capillary retention from adsorption-controlled retention.

Fig. 9(b) further compares the degree of saturation (S_r) as a function of matric suction. Across the entire suction range, X-T consistently maintained the highest S_r , followed by X, U-T, and U. At 100 kPa suction, S_r increased from 17.8% in U to 27.7% in U-T, 68.7% in X, and 74.1% in X-T, corresponding to relative improvements of 56.1%, 286.9%, and 317.6%, respectively. These results highlight the dominant role of biopolymer in sustaining liquid-phase connectivity under drying conditions, while vegetation contributed additional retention through root-mediated pore reinforcement and organic inputs. Overall, the combined biopolymer-vegetation system created a more stable pore network that enhanced both saturated storage and residual moisture retention, thereby improving soil hydraulic resilience during desaturation.

3.4. Soil saturated hydraulic conductivity

Fig. 10 presents the measured saturated hydraulic conductivity (k_s) of the four soil treatments. Unamended soil (U) exhibited the highest hydraulic conductivity ($k_s = 7.4 \times 10^{-7}$ m/s), reflecting an open pore structure with limited resistance to advective water flow. The introduction of vegetation (U-T) reduced k_s to 3.6×10^{-7} m/s, corresponding to a decrease of approximately 51.6% relative to U. This reduction is consistent with previous findings showing that root development and associated organic inputs can modify pore geometry and connectivity,

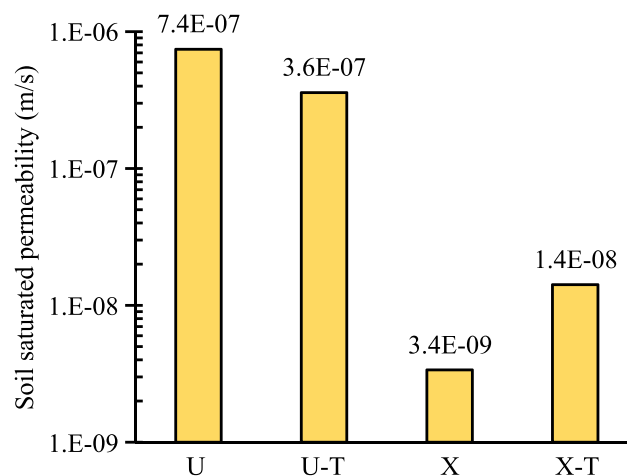


Fig. 10. Saturated hydraulic conductivity of untreated and xanthan-amended soils, with and without tall fescue vegetation. Abbreviations are the same as those in Fig. 8.

thereby lowering soil hydraulic conductivity (Lu et al., 2020). In the present study, the decline in k_s is likely attributable to root-induced pore infilling and localized compaction within the soil matrix, which increased flow tortuosity and hydraulic resistance.

The addition of xanthan gum exerted a substantially stronger influence on hydraulic conductivity. The biopolymer amended soil (X) exhibited the lowest k_s ($k_s = 3.4 \times 10^{-9}$ m/s), nearly two orders of magnitude lower than that of U. This pronounced reduction aligns with earlier studies demonstrating that microbial polysaccharides promote soil aggregation and partial pore blockage, which diminishes effective pore connectivity and restricts water transmission (Che et al., 2025). These structural modifications collectively led to a marked decline in saturated hydraulic conductivity. The combined biopolymer-vegetation treatment yielded an intermediate saturated hydraulic conductivity ($k_s = 1.4 \times 10^{-8}$ m/s), which was approximately four times higher than that of X. However, it remained much lower than U and U-T. This result indicates that xanthan gum still dominated the reduction in saturated hydraulic conductivity, whereas root growth only partially counteracted the pore-clogging effect of the biopolymer gel. The strong reduction of k_s in X was mainly caused by xanthan-induced pore sealing and increased

flow tortuosity. In X-T, by contrast, root penetration likely generated root channels, root-soil interfacial gaps, and localized macropores within the biopolymer-modified matrix. These root-associated structural features can act as preferential flow domains under saturated conditions and provide hydraulically efficient pathways that bypass part of the polymer-blocked pore system, rather than uniformly reopening the whole pore network (Lu et al., 2020). Similar behavior has been widely reported in rooted fine-grained soils, where root growth increases local pore connectivity and promotes preferential flow, even when the bulk soil matrix remains relatively dense or hydraulically restricted (Xiao et al., 2024; Cui et al., 2024). Consequently, the increase in k_s from X to X-T is best interpreted as a local connectivity effect induced by roots superimposed on a still strongly constrained biopolymer matrix.

Overall, the results suggest that biopolymer amendment is the dominant factor controlling reductions in saturated hydraulic conductivity, whereas vegetation primarily modifies flow pathways through structural reorganization rather than uniform pore constriction. The combined treatment produced a functionally balanced hydraulic state, substantially less permeable than unamended soil, yet not as restrictive as biopolymer treatment alone.

3.5. Disintegration resistance

Fig. 11 depicts the temporal evolution of soil disintegration under different treatments. U underwent rapid structural collapse upon immersion, with the disintegration rate exceeding 95% within 2 min, indicating near-complete breakdown and extremely poor resistance to water-induced degradation. In contrast, the treated soils (U-T, X, and X-T) exhibited substantially lower disintegration rates, characterized by an initial gradual increase followed by a clear stabilization phase. None of the improved soils experienced complete disintegration, and their final disintegration rates were markedly lower than that of U, confirming the significant enhancement of anti-disintegration performance by both vegetation and biopolymer amendment. Quantitatively, X displayed the highest final disintegration rate among the improved treatments (32.1%), followed by U-T (approximately 12.9%), whereas X-T showed the lowest value (6.7%), indicating the best preservation of structural integrity after prolonged immersion. These results suggest that vegetation alone provided greater resistance to water-induced disintegration than biopolymer alone, while the combined biopolymer-vegetation treatment achieved the most effective stabilization.

The contrasting behaviors can be explained by differences in bonding mechanisms and structural control. Upon immersion, U rapidly loses

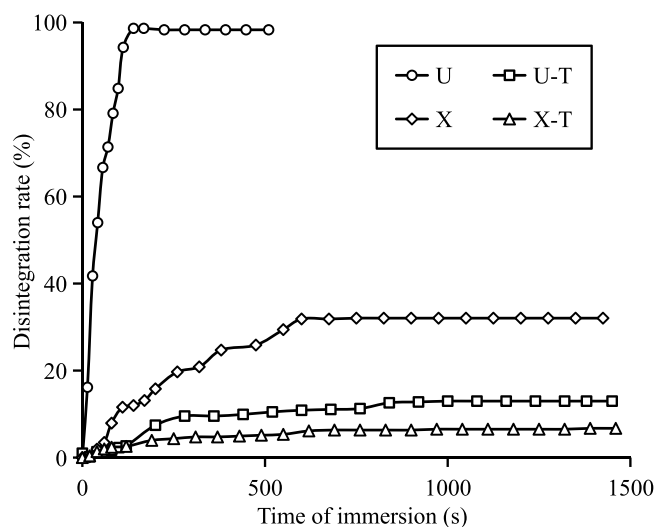


Fig. 11. Soil disintegration under different conditions. Abbreviations are the same as those in Fig. 8.

matric suction, leading to a sharp reduction in interparticle forces and immediate structural collapse. In treated soils, disintegration proceeded more gradually before reaching a stable state, reflecting the presence of additional stabilizing mechanisms. Biopolymer amendment primarily enhanced particle-scale cohesion through the formation of a three-dimensional gel-like network that bridged adjacent particles and aggregates, thereby restraining dispersion and reducing the disintegration rate. Vegetation treatment, by contrast, introduced a macroscale reinforcing framework via root entanglement, root-soil adhesion, and frictional resistance along root surfaces. This root network limited the relative movement and rotation of soil aggregates, effectively constraining structural breakdown and providing stronger resistance to disintegration than biopolymer treatment alone, consistent with previous findings (De Baets et al., 2008). The superior performance of the X-T treatment reflects complementary interactions across multiple spatial scales, biopolymer bonding enhanced particle-level cohesion, while the root system supplied structural confinement and tensile resistance at the macroscale. Together, these mechanisms created a multi-tiered stabilizing framework that resisted water-induced erosion, preserved pore network integrity, and substantially improved soil durability and resistance to water-induced structural breakdown.

3.6. Shear strength

Fig. 12 presents the shear stress-displacement responses of the four soil treatments under saturated conditions. All specimens exhibited a monotonic increase in shear stress with displacement without a distinct peak, indicating a predominantly strain-hardening response. Such behavior is typical of fine-grained soils tested under saturated conditions, where shear resistance is progressively mobilized through particle rearrangement, fabric reorientation, and interlocking rather than abrupt brittle failure. Across all applied normal stresses (25, 50, and 100 kPa), U consistently exhibited the lowest shear strength. The incorporation of vegetation (U-T) and biopolymer (X) resulted in clear strength enhancement, with increases of 32.8–59.1% and 18.4–42.3%, respectively, relative to U. Vegetation provided a comparatively stronger reinforcement effect under saturated conditions. This can be attributed to root-induced mechanical mechanisms, including root-soil interlocking, increased interface friction, and tensile resistance mobilized along the shear plane during deformation. In contrast, biopolymer primarily enhanced shear resistance through interparticle bonding and aggregation, driven by gel formation and viscous polymer bridges that improved particle connectivity. The combined biopolymer-vegetation treatment produced the greatest strength improvement, with increases of 95.1%, 55.4%, and 53.6% under normal stresses of 25, 50, and 100 kPa, respectively. This superior performance reflects the complementary roles of biopolymer-induced aggregation and root-mediated mechanical reinforcement, which together enhance interparticle continuity, restrict particle rearrangement, and promote a more stable soil skeleton during saturated shearing.

Fig. 13 illustrates the shear stress-displacement responses under an initial matric suction of 200 kPa. Compared with the saturated condition, all soils exhibited substantially higher shear strength, demonstrating the significant contribution of suction-induced apparent cohesion. U displayed a distinct peak followed by a slight post-peak reduction in shear stress, indicating mild strain-softening behavior. This response can be attributed to the increase in matric suction, which enhances interparticle capillary forces and apparent cohesion, leading to a more brittle failure mode once the peak strength is exceeded. Such suction-induced brittleness has been widely reported for unsaturated fine-grained soils, where bond breakage and collapse of capillary menisci occur during shearing (Ng et al., 2020). In contrast, U-T showed a predominantly strain-hardening response, with shear stress gradually approaching a plateau. This behavior suggests that root reinforcement effectively mitigates suction-induced brittleness by maintaining structural continuity through root-soil interlocking and tensile resistance,

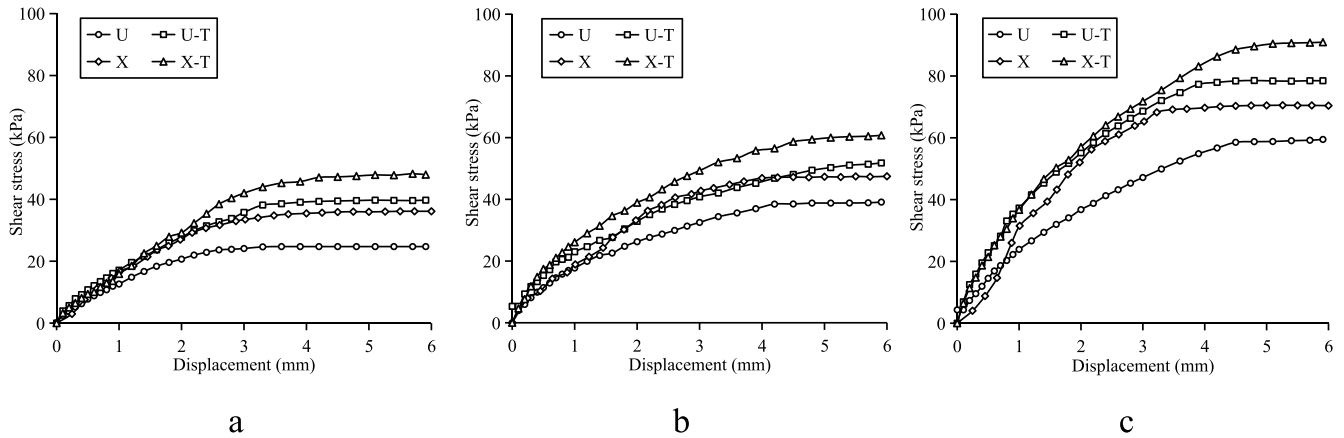


Fig. 12. Shear stress-displacement curves at saturated state under the normal stress of (a) 25 kPa; (b) 50 kPa; (c) 100 kPa. Abbreviations are the same as those in Fig. 8.

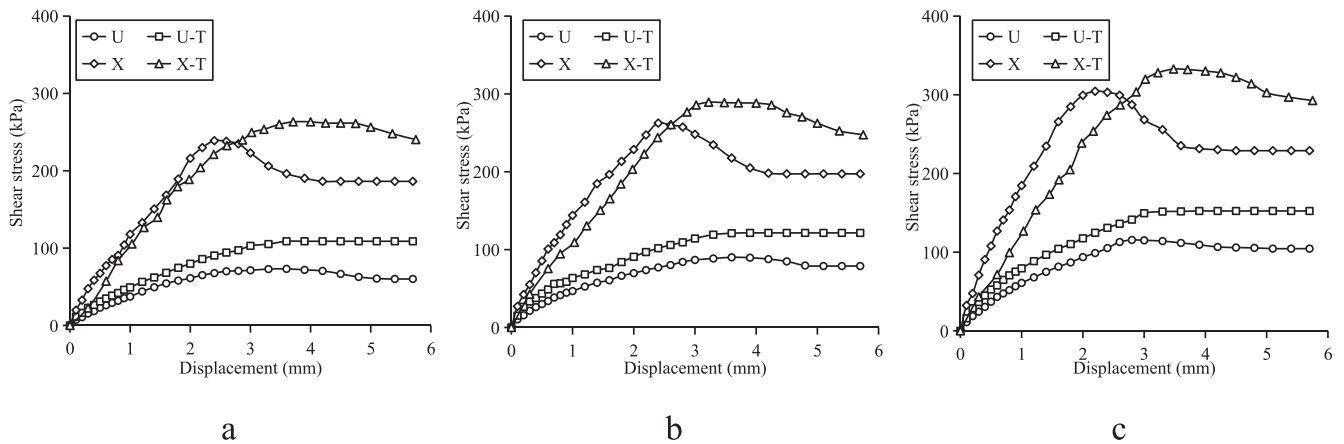


Fig. 13. Shear stress-displacement curves at 200 kPa matrix suction under the normal stress of (a) 25 kPa; (b) 50 kPa; (c) 100 kPa. Abbreviations are the same as those in Fig. 8.

thereby promoting a more ductile shear response under unsaturated conditions. Both X and X-T exhibited typical strain-softening behavior characterized by a pronounced peak followed by a reduction toward a residual value. The high peak strength originates from strong biopolymer-induced bonding and aggregation under elevated suction, which substantially increases apparent cohesion. Subsequent softening reflects progressive debonding and rupture of polymer bridges during shear deformation. Notably, the drop from peak to residual strength was less pronounced in X-T than in X, indicating that root reinforcement helps moderate brittleness and preserve post-peak structural integrity.

Increasing matric suction from saturation to 200 kPa led to substantial shear strength enhancement across all treatments, with increases of 94.9–196.4% for U, 92.6–176.8% for U-T, 334.4–574.9% for X, and 266.0–446.3% for X-T. These results demonstrate that suction-induced strengthening is markedly more pronounced in biopolymer amended soils than in vegetation-reinforced soils, owing to the strong coupling between matric suction and polymer-induced bonding. At 200 kPa suction, vegetation and biopolymer treatments increased shear strength by 31.3–48.5% and 163.8–226.0%, respectively, relative to U, indicating that at high suction levels the contribution of biopolymer clearly exceeds that of vegetation. The X-T treatment again yielded the greatest overall improvement, with shear strength increases of 259.7%, 221.1%, and 188.3% under normal stresses of 25, 50, and 100 kPa, respectively. This pronounced enhancement highlights the synergistic interaction between suction-dependent biopolymer bonding and root-induced mechanical reinforcement under unsaturated conditions.

The shear strength parameters under different treatments were determined using the total stress approach. Fig. 14 summarizes the variations in cohesion and internal friction angle for both saturated and unsaturated conditions. Under fully saturated conditions, both biopolymer and vegetation treatments substantially increased soil cohesion, whereas their influence on the internal friction angle was comparatively limited. Relative to U, cohesion increased by 89.6%, 75.1%, and 142.7% for U-T, X, and X-T, respectively, while the corresponding increases in friction angle were only 2.1%, 13.9%, and 23.1%. These results indicate that, under saturation, shear strength enhancement was governed primarily by improved interparticle bonding rather than by changes in frictional resistance. Vegetation contributed to cohesion through root wrapping, mechanical interlocking, and organic cementation associated with root exudates, which collectively strengthened particle contacts and improved structural continuity (Stokes et al., 2009). Meanwhile, the biopolymer formed a cohesive gel network that enhanced adhesion between soil particles and promoted aggregation. The combined biopolymer-vegetation treatment generated the highest cohesion, reflecting the complementary action of polymer bonding and root reinforcement in stabilizing the soil skeleton.

Fig. 14(b) presents the shear strength parameters under an initial matric suction of 200 kPa. Both apparent cohesion and apparent friction angle increased markedly with suction. This enhancement is mainly attributed to suction-induced capillary forces and the associated increase in interparticle contact forces, which promote aggregation and strengthen the load-bearing soil framework. The substantial rise in

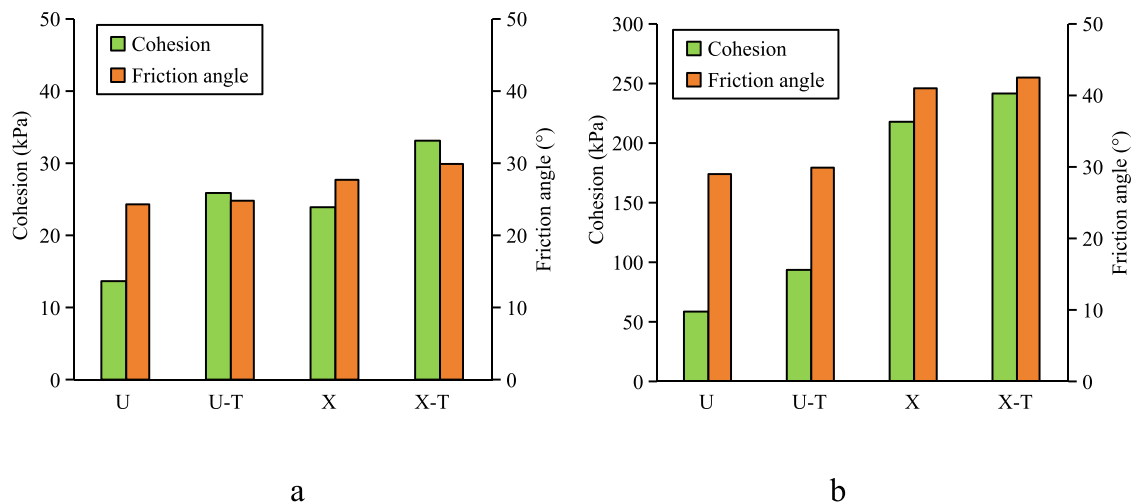


Fig. 14. Shear strength parameters of soil under the matric suction of (a) 0 kPa and (b) 200 kPa. Abbreviations are the same as those in Fig. 8.

apparent cohesion is consistent with the stress–displacement responses, in which soils under high suction exhibited stronger interparticle bonding and a greater tendency toward strain-softening behavior. Under 200 kPa suction, the cohesion of X and X-T increased sharply from 58.6 kPa in U to 217.9 kPa and 241.6 kPa, respectively. By contrast, under saturated conditions, the corresponding cohesion increments were limited to only 10.3 kPa and 19.5 kPa. This comparison clearly demonstrates that biopolymer reinforcement is far more effective under unsaturated than saturated conditions. The pronounced increase in apparent cohesion at high suction is primarily attributed to dehydration-induced hardening of the biopolymer gel network. As pore water is progressively removed, the hydrogel shrinks volumetrically and transitions from a soft, viscous state to a stiff, glassy matrix. This transformation promotes the formation of strong, tension-resistant polymer bridges between soil particles, markedly enhancing interparticle bonding compared with the swollen state under saturation (Chang et al., 2015). At the macroscopic scale, this mechanism manifests as higher peak shear strength and a shift from strain-hardening toward strain-softening behavior. In addition, plant roots embedded within the polymer-stiffened matrix provide tensile resistance and mechanical anchorage, facilitating more effective stress transfer across the shear plane. The interaction between suction-dependent polymer bonding and root reinforcement explains the slightly higher apparent cohesion observed in X-T relative to X under high suction conditions.

4. Discussion

4.1. Evaporation and water-retention regulation

The evaporation responses observed in this study demonstrate that biopolymer amendment and vegetation jointly altered soil hydraulic behavior through modifications of pore structure, water connectivity, and plant-soil interactions. The suppressed early-stage evaporation in xanthan-amended soils (X and X-T) indicates that the biopolymer network restricted rapid upward liquid flow in the near-surface zone, thereby limiting immediate atmospheric water loss. This effect reflects the ability of xanthan gum to increase flow resistance within the pore network by partially blocking macropores and enhancing microscale water retention. The markedly prolonged constant-rate stage (CRS) in X and X-T further highlights the role of xanthan gum in maintaining capillary continuity at intermediate water contents. By stabilizing liquid bridges and preserving connected flow pathways, the biopolymer delayed the transition from capillarity-dominated to diffusion-controlled evaporation, thereby sustaining upward water transport for

a longer period, consistent with previous findings on polymer hydrogels (Muguda et al., 2017). Vegetated soils (U-T and X-T) exhibited a more gradual decline in evaporation during the falling-rate stage (FRS), reflecting the combined effects of plant transpiration and root-mediated water redistribution. Root systems extract water from deeper layers, while stomatal regulation maintains relatively stable transpiration rates, resulting in smoother temporal changes in surface evaporation. This response is consistent with earlier observations that vegetation can prolong the FRS by coupling soil moisture dynamics with plant water uptake (Ng et al., 2016).

Among all treatments, the X-T composite soil displayed the most favorable hydraulic performance. The sustained evaporation plateau suggests a synergistic interaction between biopolymer and vegetation: the xanthan hydrogel functions as a moisture reservoir that alleviates plant water stress, while continuous root water uptake maintains upward hydraulic gradients and reinforces the biopolymer-stabilized capillary network. Consequently, the composite system preserves higher levels of plant-available water even in the residual stage (RS), which is critical for vegetation survival under drought conditions. These evaporation patterns are consistent with the enhanced water retention capacity measured in amended soils. In biopolymer amended soils, increases in both saturated and residual water contents stem from the strong hydration capacity and porous structure of xanthan gum, as well as its promotion of fine-particle aggregation, which enlarges the volume of water retentive pores (Zhang et al., 2022). Vegetation further modifies soil fabric through root growth and exudate secretion. Root occupation reduces effective macroporosity while increasing the proportion of smaller, water-retentive pores, and root exudates promote particle adhesion and aggregation (Ni et al., 2019). Notably, the substantial rise in residual water content in biopolymer amended soils reflects the strong adsorption and hydration capacity of the polymer matrix, which retains water even under high suction. Vegetation contributes additional residual moisture through persistent water films at the root-soil interface and the adsorption capacity of root-derived organic matter (Gao et al., 2018). Together, these mechanisms explain the superior drought resilience of the X-T treatment during repeated wetting-drying cycles.

These results also indicate that the evaporation response of the vegetated treatments was controlled not only by hydraulic water loss, but also by vegetation-induced changes in surface energy input. Canopy shading can reduce soil heating and direct evaporation, whereas root water uptake and transpiration promote water depletion from within the soil profile. The net response therefore reflects a coupled thermo-hydraulic process governed by the balance between reduced surface evaporation and increased subsurface water extraction (Badakhshan

and Vaunat, 2025). Previous studies have shown that vegetation can modify soil moisture evolution through the combined effects of shading and root-zone water uptake under changing environmental conditions (Badakhshan et al., 2025). Although this coupling was not modeled explicitly in the present study, it provides a useful framework for interpreting the distinct evaporation behavior of the X-T treatment and for future analysis of long-term moisture stability.

4.2. Pore-structure effects on hydraulic conductivity and stability

Changes in water retention are closely linked to variations in saturated hydraulic conductivity (k_s). In U-T, root growth and associated exudates partially fill and restructure large pores, reducing the number of hydraulically effective flow channels and increasing flow tortuosity, a behavior widely documented in vegetated soils (Leung et al., 2015). In X, the pronounced reduction in k_s is primarily driven by the formation of a three-dimensional xanthan gum gel network that seals interconnected pores, fills microcracks, and densifies the soil skeleton, forcing water to move through more tortuous microscale pathways. However, the X-T exhibited a slightly higher k_s than X. This can be attributed to the development of biologically induced pores and root channels within the polymer-stabilized matrix during plant growth. These preferential pathways partially counteract polymer-induced pore blockage and provide localized conduits for water flow. Consequently, the interaction between biopolymer and vegetation is synergistic with respect to water retention but partially antagonistic in terms of hydraulic conductivity reduction. The hydraulic performance of the biopolymer-root composite soil is therefore governed by the net balance between polymer-induced pore sealing and root-induced preferential flow formation.

The observed variations in k_s reveal how biopolymer amendment and vegetation jointly reshape soil pore architecture and hydraulic connectivity. These pore-scale changes exert first-order control not only on water flow pathways but also on soil stability during wetting. In unamended soil, rapid loss of matric suction within an unreinforced pore network triggers aggregate slaking and structural collapse. In contrast, amended soils display only limited initial disintegration followed by stabilization, indicating the persistence of reinforcement mechanisms beyond suction dissipation. Biopolymers enhance disintegration resistance by increasing microscale cohesion through hydrogel bonding, while vegetation provides macroscale stabilization via root entanglement, interface friction, and organic bonding. The combined X-T treatment exhibits the greatest resistance, reflecting a multiscale reinforcement mechanism in which polymer-induced particle bonding and root-mediated anchorage act jointly to suppress water-induced structural breakdown. This coupling effectively converts hydraulic stabilization into enhanced mechanical durability.

Beyond their effect on hydraulic conductivity, these pore-structure changes may also affect soil deformability and hydraulic hysteresis. Biopolymer bonding and pore sealing can change the continuity of water-filled pores during wetting and drying, while root growth and pore rearrangement may alter the stiffness and deformation response of the near-surface soil (Liu et al., 2019). Previous studies on deformable unsaturated soils have shown that water-retention hysteresis is influenced not only by wetting-drying history, but also by void-ratio change and pore-structure evolution (Badakhshan et al., 2025). Although these coupled effects were not quantified in the present study, the observed changes in evaporation, water retention, hydraulic conductivity, and shear behavior suggest that pore evolution may provide a shared structural basis for both hydraulic and mechanical responses. Future work should establish quantitative relationships among pore evolution, deformability, and hydraulic hysteresis in biopolymer-vegetation treated soils, and develop quantitative relationships or predictive models to describe these coupled effects.

4.3. Suction-dependent mechanical reinforcement

Beyond hydraulic stability, these reinforcement mechanisms also govern how matric suction is mobilized into shear resistance under unsaturated conditions. To further interpret the contribution of matric suction to shear strength, the unsaturated shear strength framework proposed by Fredlund et al. (1978) was adopted

$$\tau = c' + (\sigma - u_a)\tan\phi' + (u_a - u_w)\tan\phi^b$$

In this formulation, shear strength is expressed as the combined contribution of net normal stress and matric suction, with pore-air pressure assumed negligible and matric suction (ψ) taken as the sole suction component. Based on the direct shear results at $\psi = 0$ and 200 kPa, the suction-related parameter (ϕ^b) was evaluated, and the results are shown in Fig. 15. Here, ϕ' was assumed to remain constant and equal to that under saturated conditions, whereas c' was taken from the $\psi = 0$ condition. However, these parameters should not be regarded as strictly intrinsic material constants. The shear strength parameters at $\psi = 0$ and 200 kPa were independently obtained from direct shear tests and therefore represent apparent Mohr–Coulomb envelopes under different suction states. Accordingly, the ϕ^b value derived in this study should be interpreted as a secant-type equivalent strengthening parameter over the tested suction range, rather than a true friction angle associated solely with capillary effects. In conventional unsaturated soils, ϕ^b is theoretically close to ϕ' near the onset of desaturation and then tends to decrease as suction increases, owing to the progressive reduction in the continuity and area of water menisci (Vanapalli et al., 1996). This trend is consistent with the untreated soil (U), for which the calculated secant ϕ^b (12.9°) is markedly lower than ϕ' (24.3°). A more pronounced reduction has been reported for compacted Khon Kaen loess, where ϕ^b decreased from 27.32° at zero suction to 13.62° at 100 kPa (Chatchawan et al., 2019).

Compared with the untreated soil, the vegetation-treated soil (U-T) exhibits a higher secant ϕ^b , although it remains lower than its effective friction angle ϕ' (24.8°). This result suggests that, within the tested suction range of 0–200 kPa, matric suction is mobilized into shear resistance more efficiently in vegetated soil than in untreated soil. This enhancement is not attributed to a direct increase in fundamental frictional properties, but rather to the fact that suction improves the mechanical engagement between roots and the surrounding soil. Under unsaturated conditions, suction strengthens root–soil contact and interfacial confinement, thereby promoting more effective mobilization

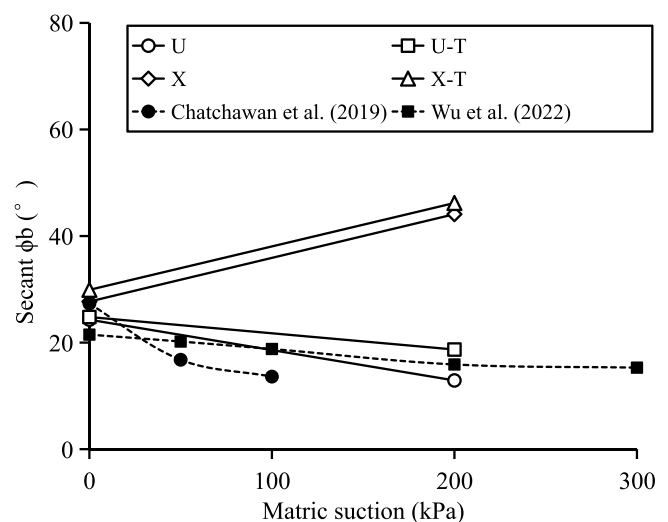


Fig. 15. Secant suction-related friction parameter ϕ^b of the tested soils under different matric suctions, with literature data for qualitative comparison. Abbreviations are the same as those in Fig. 8.

of root anchorage, interface friction, and tensile resistance across the shear zone (Badakhshan and Vaunat, 2026). A similar tendency has been reported in fiber-reinforced unsaturated clays, where fiber inclusion increased the suction-related strengthening contribution, although the efficiency of suction-induced strengthening still declined at higher suction levels (Wu et al., 2022). The corresponding literature data are included in Fig. 15 for comparison. It should be noted that the literature data in Fig. 15 are included only to illustrate the general evolution pattern of ϕ^b with suction. Because the present study was conducted on a silty clay, whereas the referenced data were obtained from other fine-grained soils, including silt and clay, the comparison should be interpreted at the level of response trend rather than strict numerical magnitude.

A fundamentally different response is observed in the biopolymer-amended soils (X and X-T), where the evaluated secant ϕ^b far exceeds the corresponding effective friction angle. This response cannot be explained by capillary effects alone, but reflects the combined contribution of suction-induced capillary stress and suction-dependent hardening of the biopolymer matrix. As matric suction increased, dehydration of the xanthan gum hydrogel likely caused volumetric shrinkage and a transition from a soft viscous gel to a stiffer, more solid-like matrix, thereby promoting stronger polymer bridges between adjacent soil particles and generating additional tensile bonding resistance (Chang et al., 2015; Guo et al., 2025). Within the tested suction interval of 0–200 kPa, the apparent contribution of matric suction to shear strength showed an enhanced trend, rather than the declining

trend commonly observed in conventional unsaturated soils. This additional resistance was primarily structural in nature and is more appropriately interpreted as an increase in apparent cohesion, rather than as a pure frictional contribution. Accordingly, the large ϕ^b values obtained for X and X-T should be understood as equivalent strengthening parameters incorporating both suction effects and polymer-induced bonding, rather than as true friction angles in the conventional sense (Khatami and O Kelly, 2013).

For the xanthan-tall fescue composite soil (X-T), the slightly higher ϕ^b relative to X indicates a further enhancement of this coupled mechanism. Roots embedded within the stiffened biopolymer-modified matrix provide additional tensile resistance and mechanical interlocking, which improve stress transfer across the shear plane and help preserve post-peak integrity. Overall, the shear strength enhancement of biopolymer-vegetation soils under unsaturated conditions is governed by a multi-scale coupled mechanism. At the microscale, suction-induced polymer hardening strengthens interparticle bonding and aggregate stability. At the macroscale, roots provide anchorage and constrain deformation. The apparent increase in suction-related strengthening parameters therefore reflects the combined effects of capillary stress, polymer bonding, and root reinforcement, rather than a simple increase in frictional resistance. Representative microstructural images (Fig. 16) further support the proposed reinforcement mechanisms. Compared with the untreated soil, the biopolymer amended soil exhibited a denser and more aggregated fabric with reduced pore openness, indicating enhanced particle bonding. In the vegetated soil, root tissue and

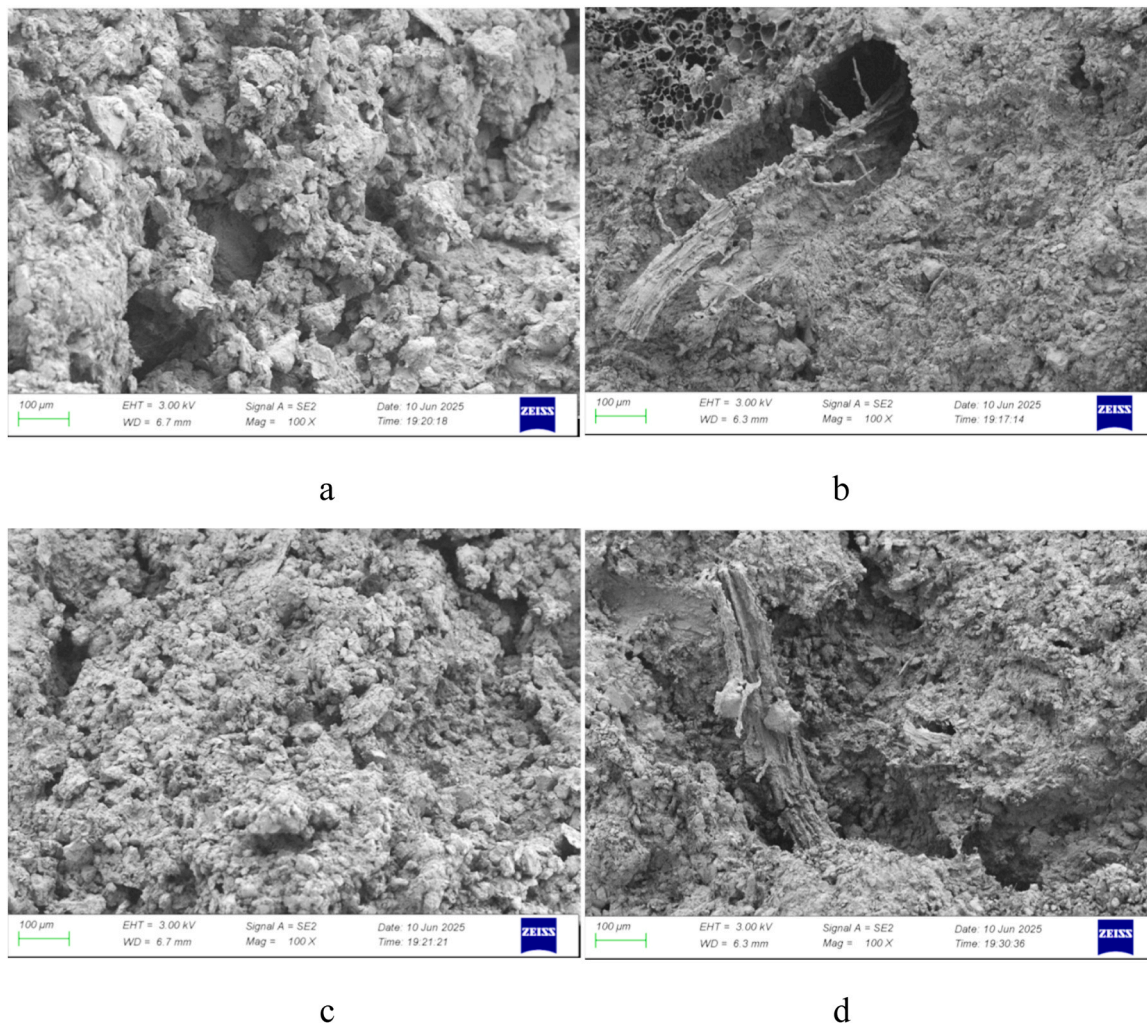


Fig. 16. Representative SEM images of (a) untreated soil; (b) xanthan amended soil; (c) vegetation-only soil; (d) xanthan-vegetation composite soil.

root-soil contact zones were clearly observed, suggesting root-induced structural rearrangement and additional anchorage within the soil matrix. In the composite treatment, the coexistence of embedded roots and biopolymer bonded aggregates indicates that particle-scale bonding and root-mediated reinforcement acted together to form a more integrated soil fabric. These microstructural features are consistent with improved water retention, lower hydraulic conductivity, greater disintegration resistance, and higher shear strength observed at the macroscopic scale.

To illustrate the coupled reinforcement mechanisms of biopolymers and plant roots under varying suction conditions, a schematic diagram is presented in Fig. 17. Under saturated conditions, biopolymer-induced particle bonding enhances apparent cohesion, while roots contribute tensile resistance and soil-root interlocking. The combination promotes synergistic shear strength improvement by improving root-soil contact. However, the biopolymer remains soft in saturated soils, limiting particle bonding and root-soil interaction, resulting in moderate strength gains. With increasing matric suction, dehydration stiffens the polymer network, strengthening both particle-particle and root-soil bonds. This enhances resistance to particle rearrangement and root pull-out, explaining the observed increase in suction-dependent friction angle for biopolymer amended soils. This behavior contrasts with conventional soils, where friction angle often plateaus or decreases at high suctions, highlighting the unique contribution of biopolymer stiffening. The biopolymer-root composite system provides multiscale reinforcement. Microscopically, polymer bonding maintains aggregate cohesion and structural integrity during wetting-drying cycles. Macroscopically, roots provide tensile anchorage and spatial confinement, limiting shear localization and disintegration. Their interaction produces consistently positive reinforcement under both saturated and unsaturated conditions.

Overall, the coupled biopolymer-vegetation system provides a bio-based approach to improve soil functional quality, promote vegetation establishment, and enhance hydro-mechanical performance during the service period relevant to ecological slope restoration. The improvement arises from the combined effects of biopolymer-induced water retention and interparticle bonding, together with root-mediated anchorage and structural modification. However, because biopolymers are inherently biodegradable, the persistence of these benefits in the field is expected to depend strongly on environmental conditions. Microbial activity can progressively decompose the biopolymer matrix and weaken the bonded soil fabric. In silty soil, the chitosan content decreased to less than half of

the initial 50 mg g⁻¹ soil after 10 days at 25 °C, whereas in sandy soil little change was observed even after 30 days (Sawaguchi et al., 2015). In addition, climatic actions such as wetting-drying and freezing-thawing cycles may accelerate strength loss through repeated swelling-shrinkage, local gel dilution, cracking, and disruption of the polymer-soil network. Previous studies have shown that freeze-thaw cycling can reduce the shear strength of biopolymer-amended soil by about 25% after five cycles (Gu et al., 2025a), while wetting-drying cycles may cause even more pronounced deterioration (Lee et al., 2022). Therefore, the present treatment should be regarded as a transitional or medium-term stabilization strategy that supports vegetation establishment and early-stage slope protection, rather than a guarantee of permanent long-term stabilization. As the biopolymer contribution gradually declines with environmental exposure, the continued development of root reinforcement may become increasingly important in sustaining the overall stability of the treated soil system. This staged reinforcement mechanism highlights the practical value of biopolymer-vegetation coupling for environmentally compatible slope restoration. Although total root biomass was measured and used as a quantitative indicator of below-ground plant development, detailed root architectural traits, including root length density, root surface area density, root diameter distribution, branching density, and root area ratio, were not quantified in this study. These parameters are more directly related to root-soil contact, preferential flow formation, and mechanical reinforcement. Therefore, this study does not establish direct quantitative relationships between root architecture and soil hydro-mechanical properties. Instead, the root-related interpretation is based on vegetation establishment, measured total root biomass, and the coupled effects of roots and xanthan gum amendment. Future studies should include image-based root morphology analysis and depth-resolved root distribution measurements to clarify the quantitative contribution of root architecture to the hydraulic and mechanical behavior of biopolymer-vegetation amended soils.

5. Conclusions

This study systematically evaluated how different biopolymers influence vegetation establishment and how biopolymer-plant interactions modify soil hydraulic behavior, structural stability, and mechanical performance. The main findings can be synthesized as follows:

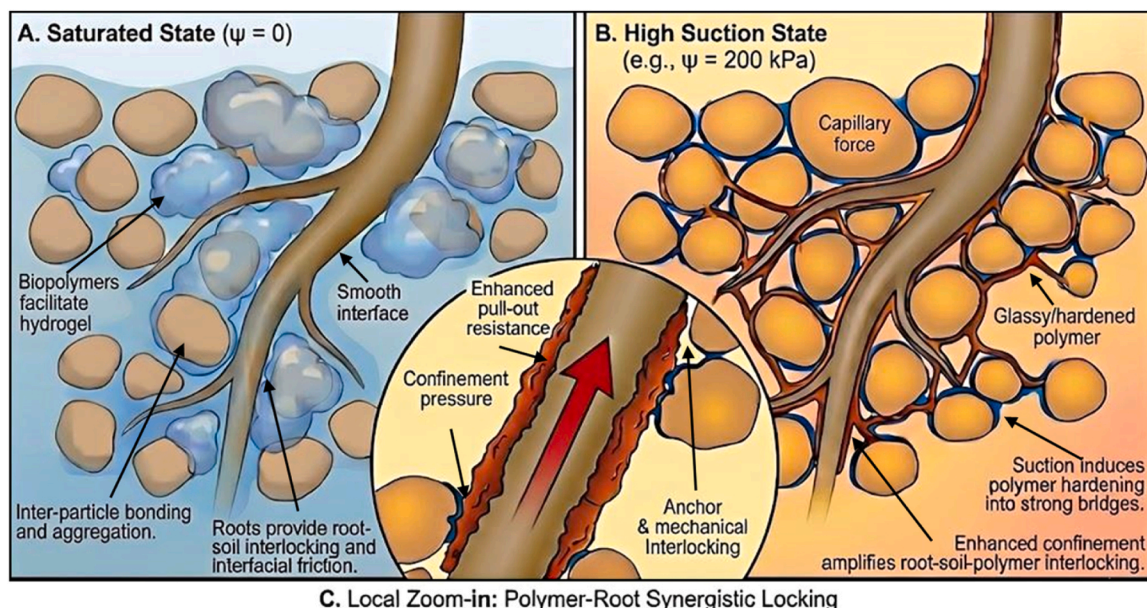


Fig. 17. Schematic diagram of the joint reinforcement mechanism of biopolymers-plant root system under different suctions.

1. Vegetation establishment was highly sensitive to biopolymer type. Among the tested species, tall fescue exhibited the highest germination rate and shoot growth performance. Xanthan gum had the least inhibitory effect on early plant growth, increasing final germination rate and shoot height by 10.0% and 10.3%, respectively, whereas chitosan delayed early seedling development.
2. Evaporation behavior of soil was substantially modified by biopolymer and vegetation. Xanthan gum reduced the initial evaporation rate from 4.29 mm/day to 2.31 mm/day and extended the constant-rate evaporation stage from about 2–5 days. During the falling-rate stage, vegetation moderated soil drying through transpiration and root-mediated redistribution of soil water, leading to a more gradual decline in surface moisture.
3. Biopolymer amendment markedly enhanced soil water retention across the full suction range. Saturated water content increased from 25.7% in unamended soil to 30.8% with xanthan gum and further to 32.2% under combined biopolymer–vegetation treatment. At 100 kPa suction, the degree of saturation of the composite soil (74.1%) was more than four times that of unamended soil (17.8%).
4. Soil saturated hydraulic conductivity (k_s) decreased from 7.4×10^{-7} m/s in unamended soil to 3.4×10^{-9} m/s after xanthan gum treatment, representing a reduction of nearly two orders of magnitude due to biopolymer-induced pore sealing. The xanthan-vegetation combined soil exhibited a slightly higher value (1.4×10^{-8} m/s), attributable to the formation of root-induced preferential flow paths that partially offset polymer-driven pore blockage.
5. Soil resistance to water-induced disintegration improved markedly. The final disintegration ratio declined from > 95% in unamended soil to 32.1% with biopolymer amendment, 12.9% with vegetation alone, and only 6.7% under combined treatment, demonstrating a strong synergistic effect of roots and biopolymer bonding on aggregate stability.
6. The reinforcing effect of biopolymer on soil shear strength was strongly suction-dependent. Under saturated conditions, vegetation and biopolymer treatments increased shear strength by 32.8–59.1% and 18.4–42.3%, respectively. At a matric suction of 200 kPa, shear strength increased by 164–226% in biopolymer amended soil, exceeding the contribution of vegetation alone (31.3–48.5%). The combined biopolymer–vegetation system exhibited the greatest enhancement, reaching 188–260%.

Overall, the coupled biopolymer-vegetation system generates a multi-scale stabilization mechanism in which biopolymer dehydration strengthens inter-particle bonds, roots modify pore architecture and water pathways, and enhanced water retention regulates soil moisture dynamics. This integration sustains water availability, prolongs evaporation plateaus, and preserves structural integrity during wetting-drying cycles. Such bio-based soil modification offers an effective and environmentally compatible approach to improve soil functional quality, facilitate plant establishment, and enhance the durability of disturbed or engineered slopes.

CRediT authorship contribution statement

Yufeng Gao: Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Junjun Ni:** Writing – review & editing, Writing – original draft, Investigation, Funding acquisition, Formal analysis. **Yanbo Chen:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Jiayu Gu:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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